Natural Language Processing basics

Intro to AI - Spring 2025 Class

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1. What is NLP?

NLP = Natural Language Processing

- Natural Language
  - Any language that occurs naturally in a human community by process of use, repetition, and change
  - **Oral*** and **written** production

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- Natural Language
  - Any language that occurs naturally in a human community by process of use, repetition, and change
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- Processing... but with what?
  - **Linguistics**: studying natural language
  - **Mathematics**: formal logic, statistics & algorithms
  - **Computer Science**: implementing algorithms and machine learning models
  - **Physics**: speech processing as wavelengths

NLP in a nutshell

The application of computational techniques for the analysis and synthesis of natural language and speech

- **Text and speech processing**: speech recognition, text-to-speech, etc.
- **Morphological analysis**: lemmatization, part-of-speech tagging, etc.

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- **Syntactic analysis**: sentence boundary disambiguation, parsing, etc.
- **Lexical semantics**: named entity recognition, sentiment analysis, etc.
- **Relational semantics**: relationship extraction, semantic parsing, etc.
- **Discourse analysis**: coreference resolution, topic segmentation, etc.

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- **Relational semantics**: relationship extraction, semantic parsing, etc.
- **Discourse analysis**: coreference resolution, topic segmentation, etc.
- **Higher level NLP applications**: summarization, machine translation, chatbots, etc.

NLP in a nutshell

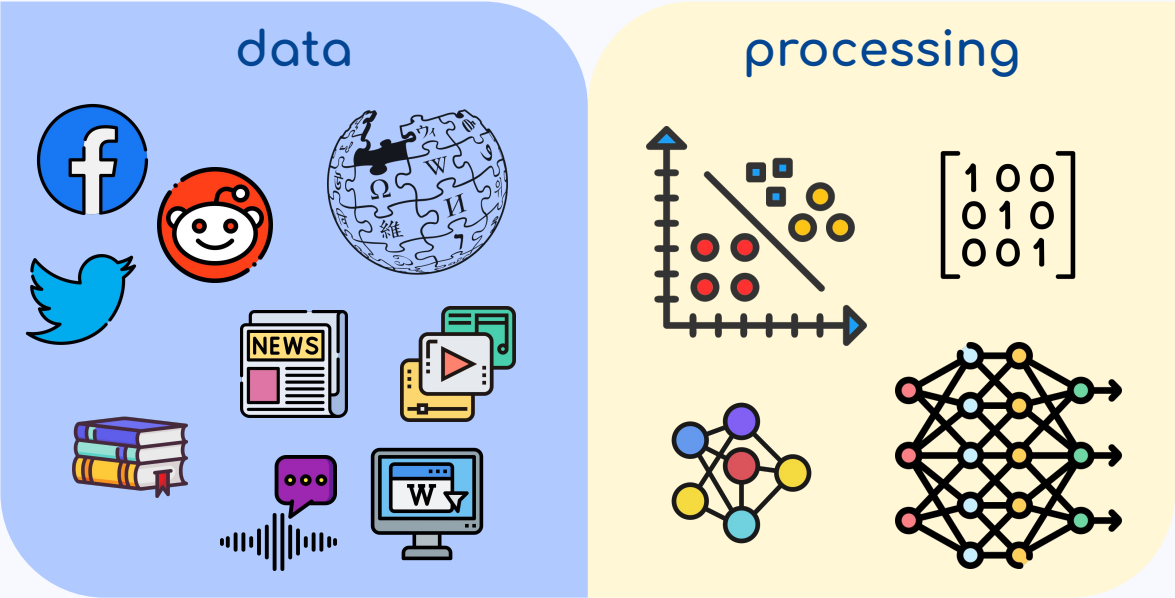
The application of **computational techniques** for the analysis and synthesis of natural language and speech

- up to 1980s: **Rule-based methods,** e.g. regular expressions, hand-written rules
  - created by experts, precise but difficult to scale
- 1990s: **Statistical methods**, e.g. linear functions
  - treating language as numerical data, deducing/learning patterns
- 2010s: **Neural networks**, a combination of linear and nonlinear functions
  - state-of-the-art performance, but harder to interpret
- 2020s: **Multi-layer, multi-head transformer neural networks**

NLP in a nutshell

Source: flaticon.com

NLP in a nutshell



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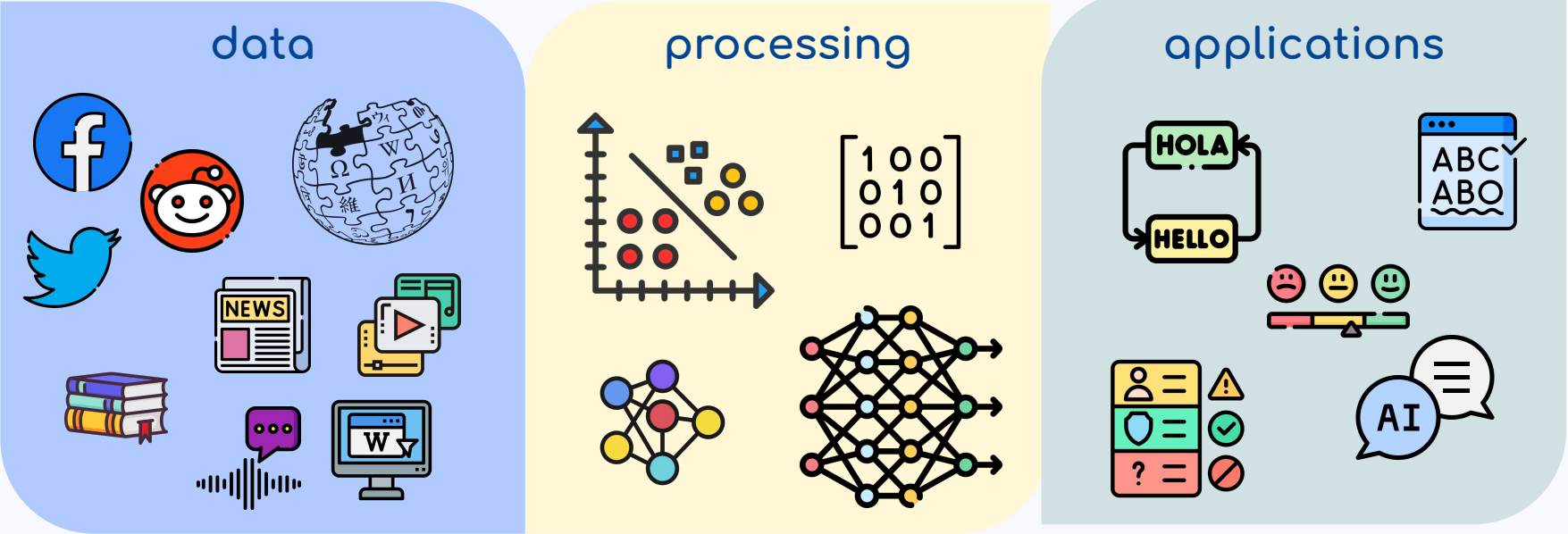

Source: flaticon.com

2. Fundamentals of linguistics

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What is linguistics?

The study of spoken & written natural language w.r.t. form, meaning, and context

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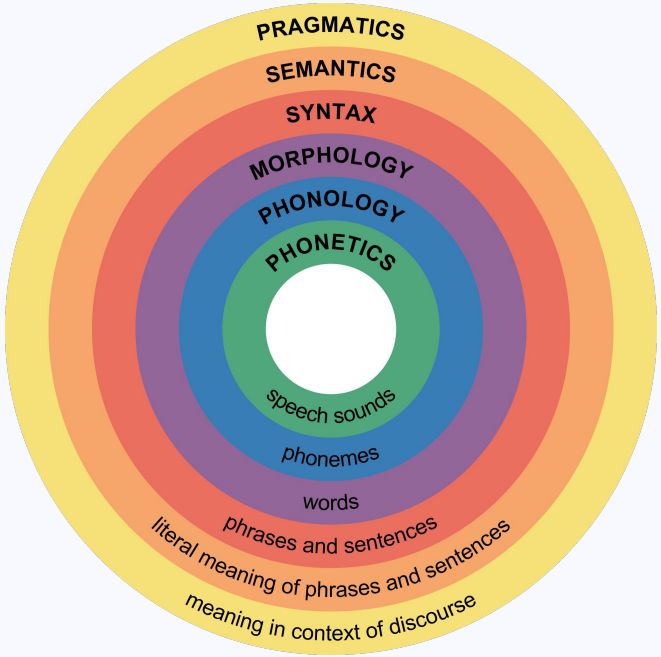
The study of spoken & written natural language w.r.t. form, meaning, and context

Phonetics:
Physical aspects of speech sounds

Phonology:
Linguistic sounds of a particular language



https://commons.wikimedia.org/wiki/File:Major_levels_of_linguistic_structure.svg

What is linguistics?

The study of spoken & written natural language w.r.t. form, meaning, and context

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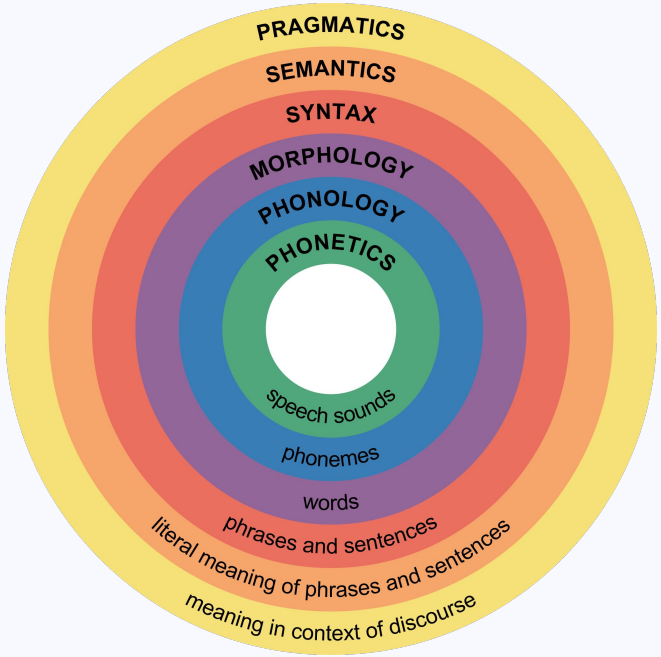
Senseful components of words and word forms



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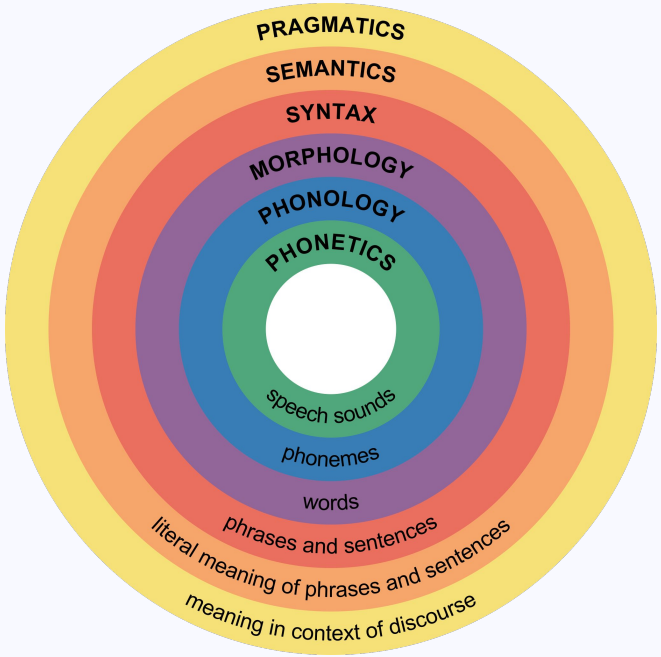

Syntax:
Structural relationships between words, usually in a sentence

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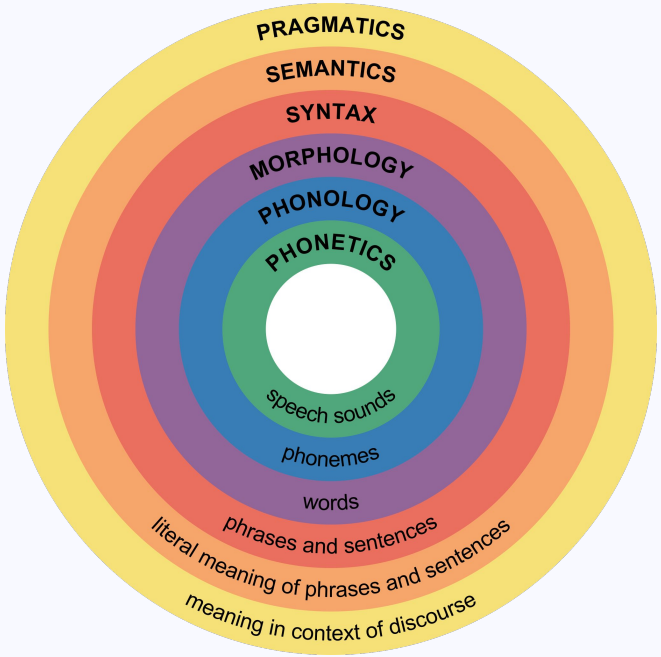

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Syntax:
Structural relationships between words, usually in a sentence

Semantics:
Meaning of single words and compositions of words

Pragmatics:
How language is used to accomplish goals

Linguistic building blocks

- **Phoneme**: Smallest unit of spoken language (usually one speech sound)

On	a	laissé	la	fenêtre	ouverte.
on	a	l ai ss é	l a	f e n ê t r e	ou v e r t e
ɔ̃	a	l e s e	l a	f ə n ɛː t ʁ ə	u v ɛ ʁ t ə

on	a	lai	ssé	la	fe	ne	tre	ou	ver	te
ɔ̃ ˈna		le	ˈse	laf		ˈnɛtʁ		u	ˈvɛʁt	

https://en.wikipedia.org/wiki/French_phonology

Linguistic building blocks

- **Morpheme**: Smallest unit with a meaning or grammatical function in both spoken and written language

ajouter (V)	
ajout	er
stem	***suffix***
meaning	*function*

ajoutent (V)	
ajout	ent
stem	***suffix***
meaning	*function*

lemma: ajouter (V)

stem: ajout-

Linguistic building blocks

- **Morpheme**: Smallest unit with a meaning or grammatical function in both spoken and written language

photocopie (N)	
photo	copie
stem	***suffix***
meaning	*function*

photocopies (N)		
photo	copi	es
stem	***stem***	***suffix***
meaning	*meaning*	*function*

photocopies (V)	
photocopi	es
stem	***suffix***
meaning	*function*

lemma: photocopie (N)
photocopier (V)

stem: photocopi-

Linguistic building blocks

- **Morpheme**: Smallest unit with a meaning or grammatical function in both spoken and written language

anticonstitutionnellement (ADV)				
anti	constitu	tion(n)	elle	ment
prefix	***stem***	***suffix***	***suffix***	***suffix***
function	*meaning*	*meaning*	*function*	*function*

lemma: anticonstitutionnellement (ADV)

stem: constitu-

Linguistic building blocks

- **Word?**
  - **Simple terms:** the string of letters between two spaces or punctuation?
    but… *pin's, sac à dos, Tour Eiffel* are one word each!
  - **Phonetics/Phonology:** the cluster of sounds between two pauses?
    but… we often pronounce *je suis* as *chui* !
  - **Morphology**: a cluster of morphemes with a distinct meaning and distinct function?
    but… if we split *langue-de-chat* 🥖 , it will become *langue de chat* ! 😸

Linguistic building blocks

- **Word**
  - The smallest unit of language that is to be uttered in isolation
  - Has either a lexical function (e.g. verbs, nouns) or a grammatical function (e.g. articles)
  - Every word is composed of one or more morpheme: (prefix) - **stem** - (suffix)

Linguistic building blocks

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Linguistic building blocks

- **Sentence?**
  - **Simple terms:** a string of words starting with a capital letter and ending in punctuation?
    but… where does "*Jules César a été assassiné en 44 av. J.-C. en plein sénat.*" start/end?
  - **Syntax:** a unit consisting of a subject and predicate?
    but… "*Métro, boulot, dodo.*" doesn't have these elements, but is a sentence!
  - **Semantics**: a cluster of words with a complete meaning?
    but… "*Yes!*" is a sentence, with complete meaning in the context of a discussion

Linguistic building blocks

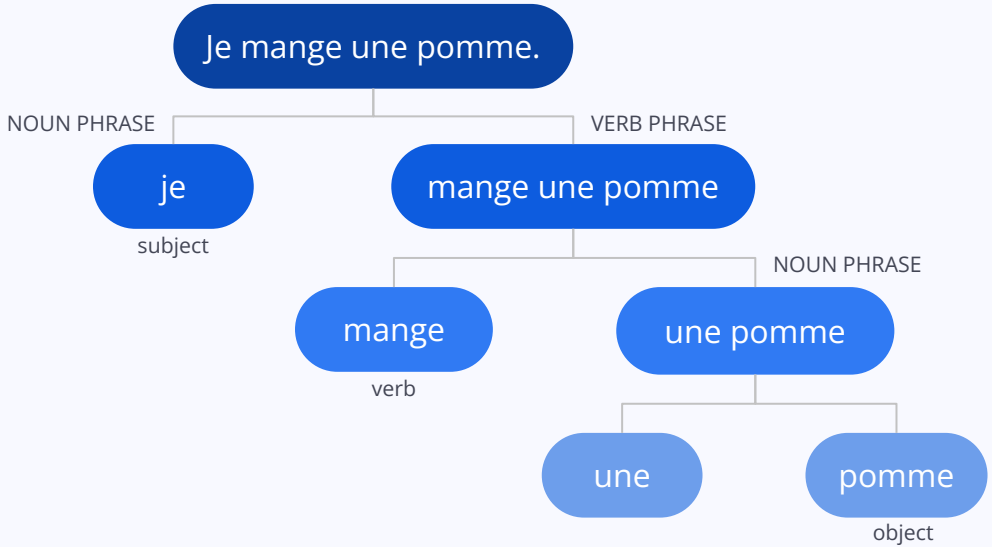
- **Sentence**: a set of words that is
  - complete in itself
  - (typically) contains a subject and predicate
  - (typically) conveys a statement, question, exclamation, or command
  - (typically) consists of a main clause and sometimes one or more subordinate clauses

- **Phases → Sentences → Clause**

Clause:	Je pense que ton chien aime Marie.				
Sentence:	je pense		(que) ton chien aime Marie		
Phrase:	je	pense	ton chien	aime	Marie

Linguistic building blocks

- **Sentence**:
  - Subject:
    - Subject
    - Articles
    - Complements, e.g. adjectives
  - Predicate:
    - Verb
    - Direct object
    - Indirect object
    - Complements, e.g. adverbs, prepositions, etc.



Linguistic building blocks

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object

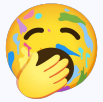
And yet… we're blocked

Even when sentences are simple… semantics are not!

- **Lexical ambiguity**:

  Je mange un avocat. ⇒ 🍽️🥑 or 🍽️👨‍⚖️ ?

- **Syntactic ambiguity**:

  L'artiste peint la nuit. ⇒  or  ?

- **Grammaticality vs. Plausibility**:

  La banane court un marathon. ⇒ 🍌🏃 ?

And yet… we're blocked

Even when sentences are simple… semantics are not!

- **Coreference**:

  Sylvain a caressé son chien et il lui a donné une friandise. ⇒ 👨‍🦰 or 🐕?
- **Pragmatic ambiguity**:

  Sylvain a vu un homme avec un télescope. ⇒ Who has the telescope?
- **Garden path sentences**:

  Sylvain a vu un homme avec un télescope, qu'il a ensuite ramené chez lui pour lui demander conseil.

Source: https://french.stackexchange.com/questions/8409/garden-path-sentence-en-fran%C3%A7ais

And yet… we’re blocked

Even when sentences are simple… semantics are not!

- **Sarcasm**:

  Oh mais j'adore faire la queue à la banque! ⇒ Wait, really?

- **Negation and contrast**:

  Mes nouvelles baskets sont moches mais tellement confortables. ⇒ ⭐ or ⭐⭐⭐⭐⭐?

And yet… we're blocked

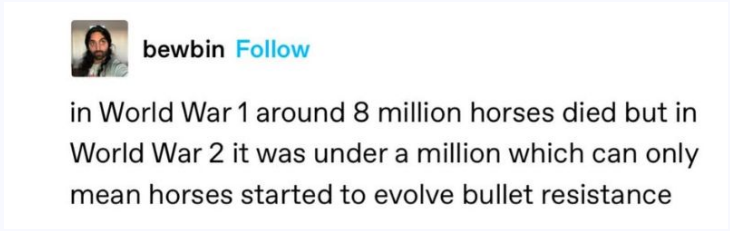
Language in grammar books ≠ Language IRL!

- **Slang, neologisms, abbreviations, etc.**

- **Spelling errors, neologisms, creativity, etc.**

https://x.com/NathInReims/status/659716375099871232

- **Inference from world knowledge**



https://bewbin.tumblr.com/

3. Feature Extraction

(Natural Language Preprocessing)

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Processing text:

- A word
  - Tokenization
  - Lemmatization/Stemming

Processing text:

- A word
  - Tokenization
  - ~~Lemmatization/Stemming~~
- A sentence
  - Lower-casing
  - Segmentation
  - Lemmatization/Stemming
  - Part-of-speech tagging
  - Stop word removal
  - Syntactic analysis

Processing text:

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  - Part-of-speech tagging
  - Stop word removal
  - Syntactic analysis
- A paragraph
  - Syntactic analysis
  - Word disambiguation
  - Named Entity Recognition
  - Coreference resolution
  - Discourse parsing

Tokenization

Separate a word to: morphemes for humans, **tokens** for machines!

- **Character tokens**:

J'écoute les oiseaux.																	
j	‘	é	c	o	u	t	e	l	e	s	o	i	s	e	a	u	x	.

- **Subword tokens**:

J'écoute les oiseaux.							
j’	écout	e	les	ois	eau	x	.

We determine these subtokens based on frequency or with algorithms (e.g. BPE)

- **Word tokens**:

J'écoute les oiseaux.					
j’	écoute	le	s	oiseaux	.

Tokenization

Vocabulary (NLP):

Set of all tokens in the data we are using

Separate a word to: morphemes for humans, **tokens** for machines!

- **Character tokens**:

J'écoute les oiseaux.																	
j	‘	é	c	o	u	t	e	l	e	s	o	i	s	e	a	u	x	.

- **Subword tokens**:

J'écoute les oiseaux.							
j’	écout	e	les	ois	eau	x	.

- **Word tokens**:

J'écoute les oiseaux.					
j’	écoute	le	s	oiseaux	.

small vocabulary
many tokens/sentence

large vocabulary
few tokens/sentence

Lower-casing, Segmentation

- Removing capitalization, splitting a sentence in words

J'écoute les oiseaux.				
j’	écoute	les	oiseaux	.

Lemmatization/Stemming

- Turning the word to its lemma and stem

J'écoute les oiseaux.				
j’	écoute	les	oiseaux	.
PRON	**VERB**	**DET**	**NOUN**	**PUNCT**

je	*écouter*	*le*	*oiseau*	.
je	*écout*	*le*	*oiseau*	.

Why not at word level?

Meaning in sentence!

Part-of-speech tagging

- Labeling words with their part-of-speech type, based on the current sentence

J'écoute les oiseaux.				
j’	écoute	les	oiseaux	.
PRON	**VERB**	**DET**	**NOUN**	**PUNCT**

Stop word removal

- Removing high-frequency words that do not contribute to sentence's unique meaning

J'écoute les oiseaux.				
j'	écoute	les	oiseaux	.
PRON	**VERB**	**DET**	**NOUN**	**PUNCT**

je	*écouter*	*le*	*oiseau*	.
je	*écout*	*le*	*oiseau*	.

~~*j'*~~	*écoute*	~~*les*~~	*oiseaux*	.

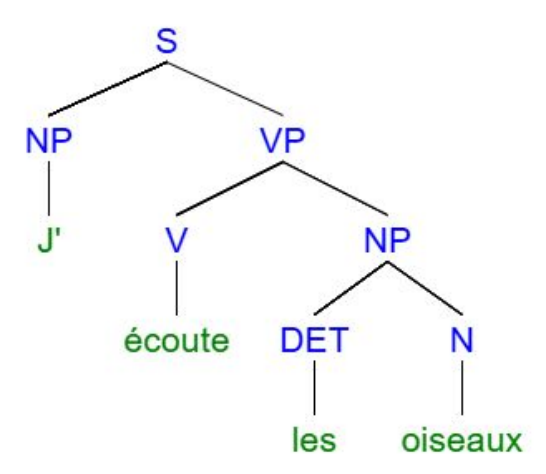
How do we find these stop words?

With statistics...

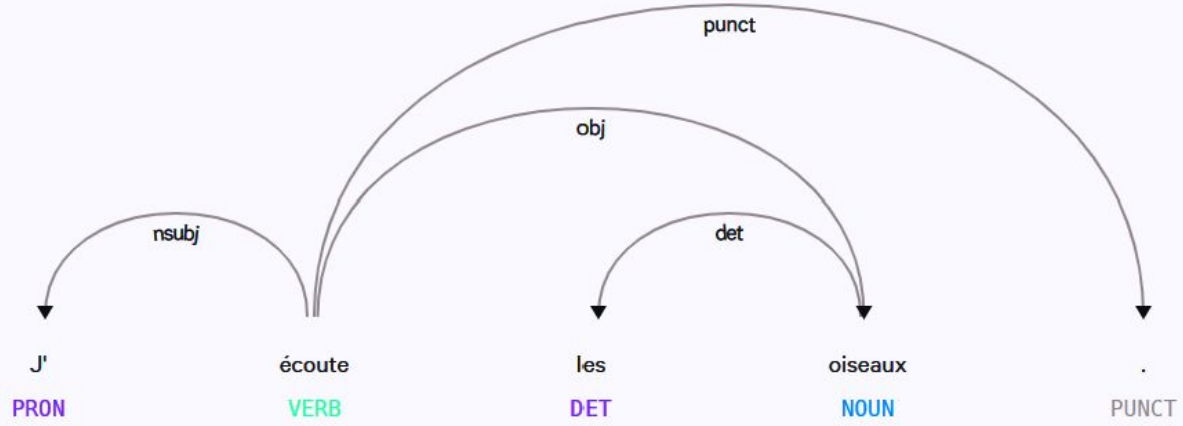
...but we can also use premade tools and lists

Syntactic parsing

- Syntactic analysis of a sentence
- **Human-readable way**: constituency graphs or dependency graphs



https://mshang.ca/syntree/



https://demos.explosion.ai/displacy

Syntactic parsing

- **Machine-readable way**: constituency parsing or dependency parsing

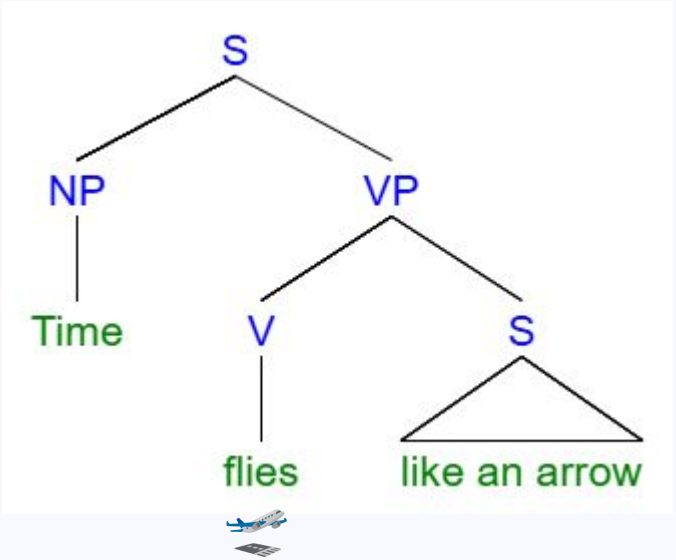
# sent_id = 1									
# text = J'écoute les oiseaux.									
1	**J'**	moi	PRON	_	Emph=No|Number=Sing|Person=1|PronType=Prs	2	nsubj	_	SpaceAfter=No|TokenRange=0:2
2	**écoute**	écouter	VERB	_	Mood=Ind|Number=Sing|Person=1|Tense=Pres|VerbForm=Fin	0	root	_	TokenRange=2:8
3	**les**	le	DET	_	Definite=Def|Number=Plur|PronType=Art	4	det	_	TokenRange=9:12
4	**oiseaux**	oiseau	NOUN	_	Gender=Masc|Number=Plur	2	obj	_	SpaceAfter=No|TokenRange=13:20
5	**.**	.	PUNCT	_	_	2	punct	_	SpaceAfter=No|TokenRange=20:21

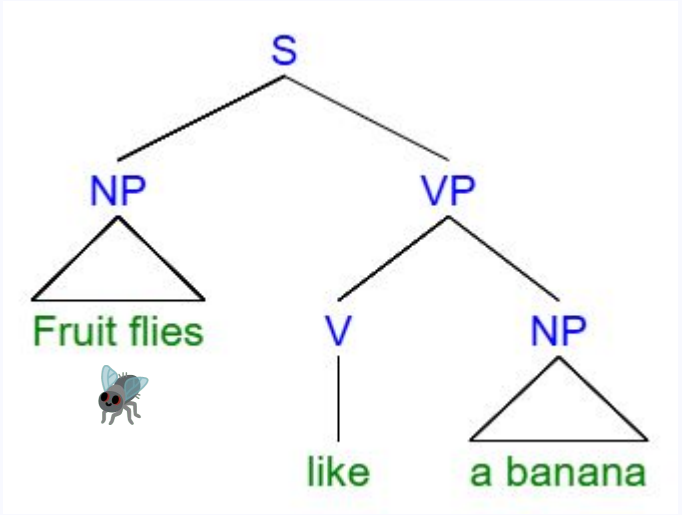
↑ **No.** ↑ **word** ↑ **lemma** ↑ **POS** ↑ **morpho-syntactic features** ↑ **non-linguistic features**

https://lindat.mff.cuni.cz/services/udpipe/

Word disambiguation

- Finding the specific meaning of a word in its context, e.g. with syntactic parsing:
  “Time flies like an arrow, fruit flies like a banana.” ⇒ is *flies* a verb or a noun?



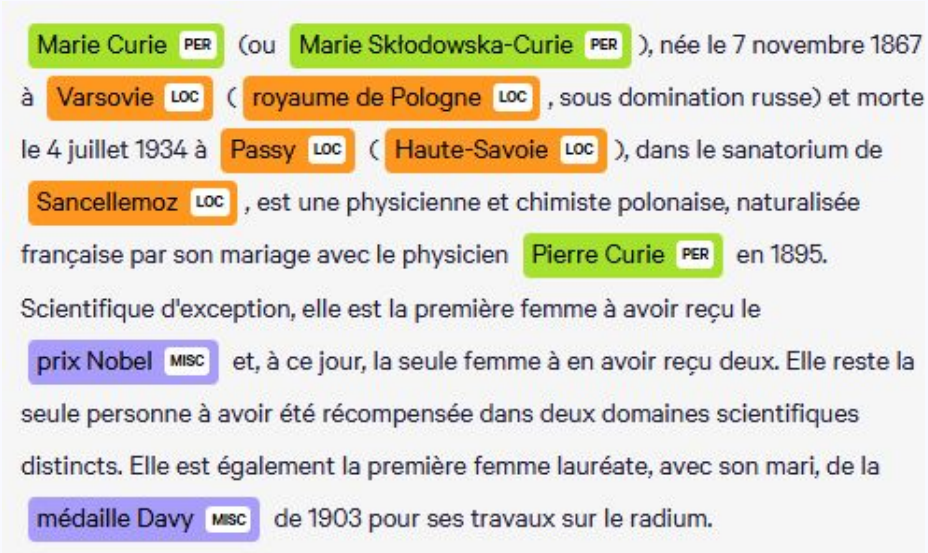


Named Entity Recognition

- Finding proper nouns and their type (people, places, companies, etc.)

Marie Curie (ou Marie Skłodowska-Curie), née le 7 novembre 1867 à Varsovie (royaume de Pologne, sous domination russe) et morte le 4 juillet 1934 à Passy (Haute-Savoie), dans le sanatorium de Sancellemoz, est une physicienne et chimiste polonaise, naturalisée française par son mariage avec le physicien Pierre Curie en 1895. Scientifique d'exception, elle est la première femme à avoir reçu le prix Nobel et, à ce jour, la seule femme à en avoir reçu deux. Elle reste la seule personne à avoir été récompensée dans deux domaines scientifiques distincts. Elle est également la première femme lauréate, avec son mari, de la médaille Davy de 1903 pour ses travaux sur le radium.

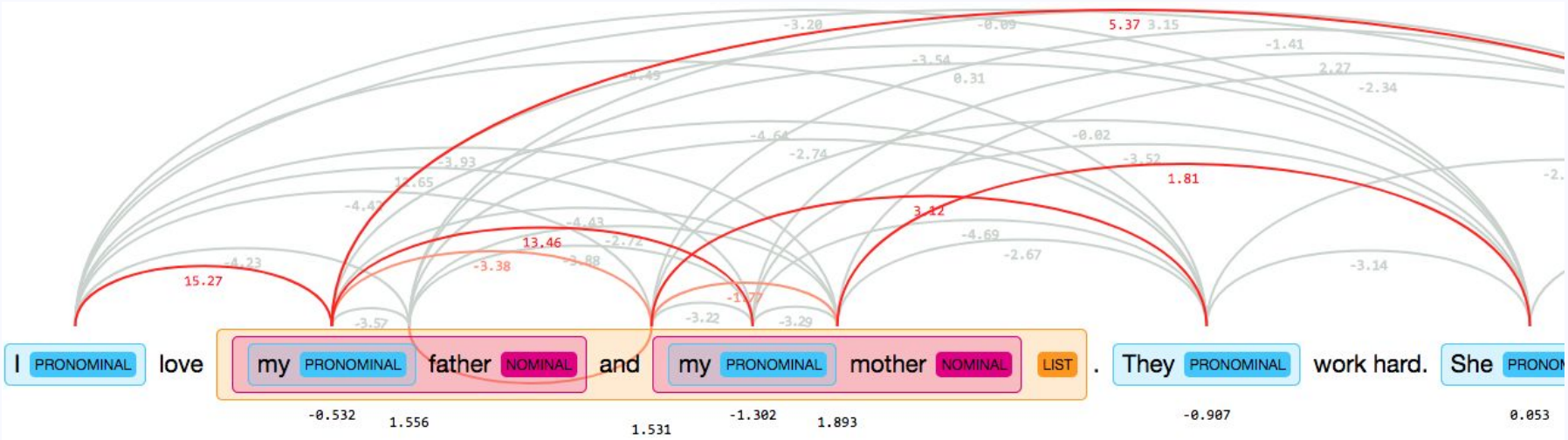
https://fr.wikipedia.org/wiki/Marie_Curie



https://demos.explosion.ai/displacy-ent

Coreference resolution

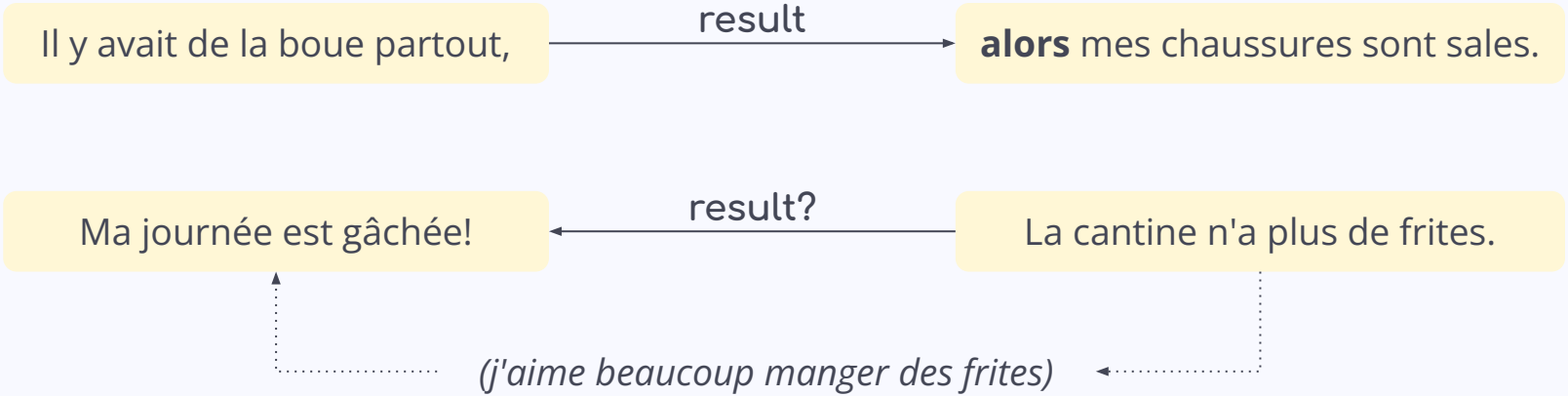
- Finding entities (people, places, concepts, etc.) that are linked together, in one or across many sentences



https://spacy.io/universe/project/neuralcoref-vizualizer

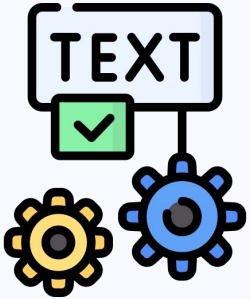
Discourse parsing

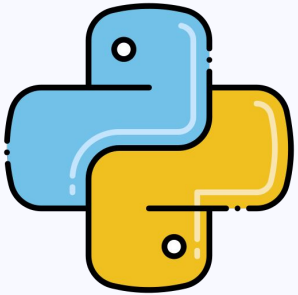
- Linking sentences or clauses with semantic-pragmatic relations:
  cause, explanation, result, contrast, etc.
- **Explicit** (with connector words) or **Implicit** (without a visible connection)



4. Getting started in NLP

4.1 Our tools for NLP





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Finding data

Plenty of resources online:

- NLTK datasets and lexical resources
- Scikit-learn datasets
- Kaggle competitions
- Linguistic Data Consortium
- Research groups, Shared Tasks, etc.
  - https://nlp.stanford.edu/links/statnlp.html#Corpora
  - http://www.llf.cnrs.fr/en/resources
- More and more

Or DIY!

- Web scraping (careful!)
  - BeautifulSoup4
- Wikimedia dumps
- Reddit dumps
- T~~witter API~~

Corpus (pl. corpora):

A dataset of text

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Tools for (pre)processing

- Natural Language Toolkit (NLTK)
  - Corpora, Preprocessing tools, Statistical methods, (Basic) High-level applications

- SpaCy
  - Preprocessing tools, Language models, Statistical & neural methods, Visualizers, High-level applications & pipelines

spaCy

- Hugging Face
  - The go-to for (large) language models, Corpora, Built-in preprocessing for (L)LMs

Tools for (pre)processing

- Gensim
  - ML Python library Statistical methods, Word embeddings!

- scikit-learn
  - ML Python library, the go-to for Statistical methods, Corpora

- Honorable mentions:
  - NumPy
  - Pandas
  - matplotlib & seaborn

Tokenization example: NLTK

```
import nltk
from nltk.tokenize import (word_tokenize,
                           sent_tokenize,
                           TreebankWordTokenizer,
                           wordpunct_tokenize,
                           TweetTokenizer,
                           MWETokenizer)
text="Hope, is the only thing stronger than fear! #Hope #Amal.M"
```

import the library

different types of tokenizers, **imported as functions**

input text

```
print(word_tokenize(text))
```

```
['Hope', ',', 'is', 'the', 'only', 'thing', 'stronger', 'than', 'fear', '!', '#', 'Hope', '#', 'Amal.M']
```

```
print(sent_tokenize(text))
```

```
['Hope, is the only thing stronger than fear!', '#Hope #Amal.M']
```

```
text="What you don't want to be done to yourself, don't do to others..."
tokenizer= TreebankWordTokenizer()
print(tokenizer.tokenize(text))
```

```
['What', 'you', 'do', "n't", 'want', 'to', 'be', 'done', 'to', 'yourself', ',', 'do', "n't", 'do', 'to', 'others',
'...']
```

Tokenization: SpaCy

import the library

```
import spacy
nlp = spacy.load("en_core_web_sm")
```

import the language model

```
doc = nlp("\"Next Week, We're coming from U.S.!\"")
for token in doc:
    print(token.text)
```

process the input text

(*doc* is an object)

```
"
Next
Week
,
We
're
coming
from
U.S.
!
"
```

https://ashutoshtripathi.com/2020/04/06/guide-to-tokenization-lemmatization-stop-words-and-phrase-matching-using-spacy/

Lemmatization, POS tagging, etc.: NLTK

```
import nltk

text = "John loves to play soccer"
tokens = nltk.word_tokenize(text)
pos_tags = nltk.pos_tag(tokens)

print(pos_tags)
```

```
[('John', 'NNP'), ('loves', 'VBZ'), ('to', 'TO'), ('play', 'VB'), ('soccer', 'NN')]
```

```
from nltk.stem import WordNetLemmatizer

text = "Hello i am Data Scientist and i love Reading books."

# Tokenize the text
tokens = nltk.word_tokenize(text)

# Lemmatize the tokens
lemmatizer = WordNetLemmatizer()
lemmatized_tokens = [lemmatizer.lemmatize(token) for token in tokens]

print(lemmatized_tokens)
```

```
['Hello', 'i', 'am', 'Data', 'Scientist', 'and', 'i', 'love', 'Reading', 'book', '.']
```

Python Reminder:

to use a function **Y()** from library **X**:

`import X`
use `X.Y()` in code

<u>or</u>

`from X import Y`
use `Y()` in code

https://www.nomidl.com/natural-language-processing/stemming-and-lemmatization/
https://www.nomidl.com/natural-language-processing/part-of-speech-tagging-and-named-entity-recognition/

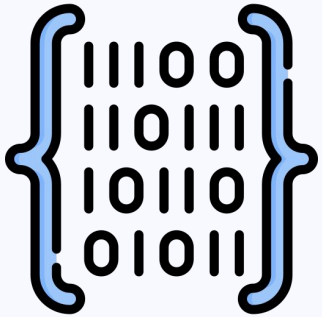
Lemmatization, POS tagging, etc.: SpaCy

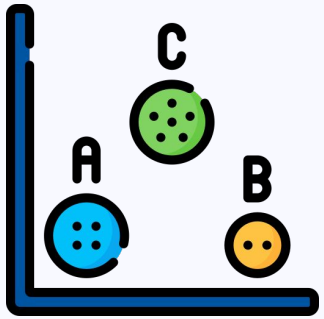
```
doc1 = nlp(u"I am a runner running in a race because I love to run since I ran today")
for token in doc1:
    print(token.text, '\t', token.pos_, '\t', token.lemma, '\t', token.lemma_)
```

```
I         PRON    561228191312463089      -PRON-
am        VERB    10382539506755952630    be
a         DET     11901859001352538922    a
runner    NOUN    12640964157389618806    runner
running           VERB    12767647472892411841    run
in        ADP     3002984154512732771     in
a         DET     11901859001352538922    a
race      NOUN    8048469955494714898     race
because           ADP     16950148841647037698    because
I         PRON    561228191312463089      -PRON-
love      VERB    3702023516439754181     love
to        PART    3791531372978436496     to
run       VERB    12767647472892411841    run
since     ADP     10066841407251338481    since
I         PRON    561228191312463089      -PRON-
ran       VERB    12767647472892411841    run
today     NOUN    11042482332948150395    today
```

https://ashutoshtripathi.com/2020/04/06/guide-to-tokenization-lemmatization-stop-words-and-phrase-matching-using-spacy/

4.2 Representing data





Human readable ≠ Machine readable

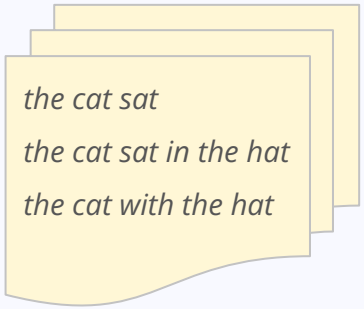
Text Encoding: Converting meaningful text into number/vector representations

One token = One element in the vector

Encoding: Bag-of-Words

- Assign **a value to each token**, based on its **presence** in each sentence wrt. corpus

1. Documents → 2. Vocabulary → 3. Count



- *the*
- *cat*
- *sat*
- *in*
- *hat*
- *with*

→ (not in corpus!)

Sentence	the	cat	sat	in	hat	with
the cat sat	1	1	1	0	0	0
the cat sat in the hat	2	1	1	1	1	0
the cat with the hat	2	1	0	0	1	1
the dog with the hat	2	0	0	0	1	1

https://victorzhou.com/blog/bag-of-words/

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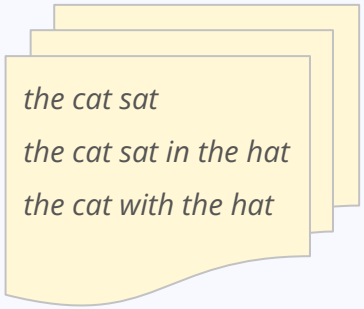

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the cat sat in the hat	2	1	1	1	1	0
the cat with the hat	2	1	0	0	1	1
the dog with the hat	2	0	0	0	1	1

↳ 4. Sentence vectors!

```
[1, 1, 1, 0, 0, 0]
[2, 1, 1, 1, 1, 0]
[2, 1, 0, 0, 1, 1]
[2, 0, 0, 0, 1, 1]
```

https://victorzhou.com/blog/bag-of-words/

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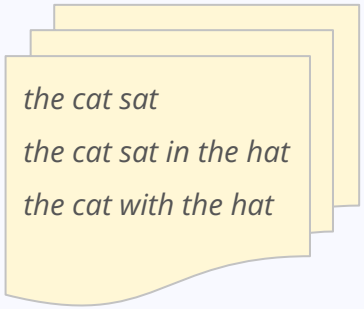

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↳ 4. Sentence vectors!

```
[1, 1, 1, 0, 0, 0]
[2, 1, 1, 1, 1, 0]
[2, 1, 0, 0, 1, 1]
[2, 0, 0, 0, 1, 1]
```

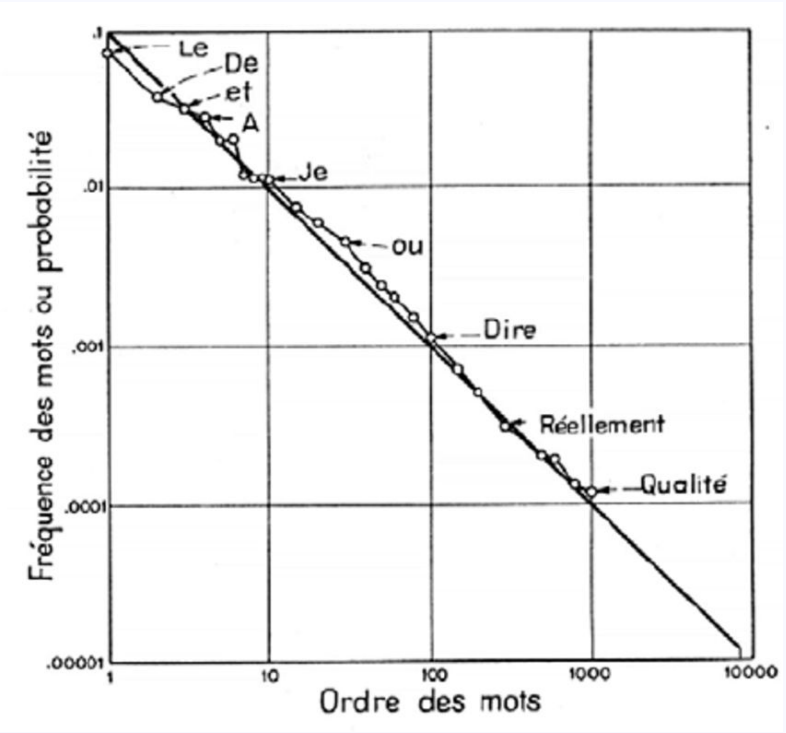
But… *cat* = *hat* !

https://victorzhou.com/blog/bag-of-words/

Frequency

- Zipf's law: in an ordered list of words (per word frequency) in a document… …the frequency of a word is inversely proportional to its rank (1st, 2nd, 500th, etc.)
- Applies to economics, characters, syllables, words, phrases, across languages!

- **Improve word encodings by adding information about word frequency**



Mandelbrot, Benoît, (1965), Forecasts of Future Prices, Unbiased Markets, and "Martingale" Models, The Journal of Business.

Frequency

- **TF-IDF**: measuring the importance of a word in a set of documents

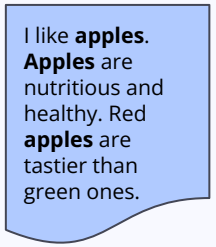
Term Frequency

$$TF = \frac{\text{number of times the term appears in the document}}{\text{total number of terms in the document}}$$

x

Inverse Document Frequency

$$IDF = log(\frac{\text{number of the documents in the corpus}}{\text{number of documents in the corpus contain the term}})$$



$$TF = \frac{3}{15} = 0.2$$

$$IDF = log\frac{10}{2} = 0.70$$

$$TF - IDF_{apples} = 0.2 * 0.70 = 0.14$$

https://www.learndatasci.com/glossary/tf-idf-term-frequency-inverse-document-frequency/

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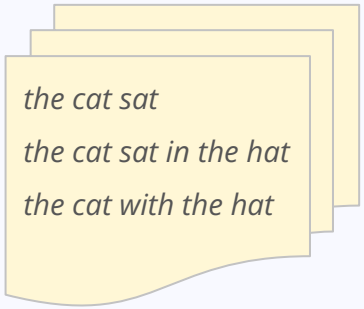

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the cat with the hat	2	1	0	0	1	1
the dog with the hat	2	0	0	0	1	1

↳ 4 + 5. Vectors with TF-IDF!

```
[0.33, 0.33, 0.33, 0, 0, 0]
[1.33, 0.17, 0.17, 0.17, 0.17, 0]
[1.6, 0.2, 0, 0, 0.2, 0.2]
[1.6, 0, 0, 0, 0.2, 0.2]
```

https://victorzhou.com/blog/bag-of-words/

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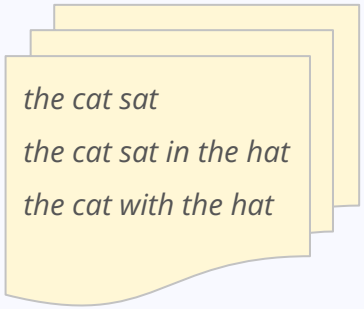

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the dog with the hat	2	0	0	0	1	1

↳ 4 + 5. Vectors with TF-IDF!

```
[0.33, 0.33, 0.33, 0, 0, 0]
[1.33, 0.17, 0.17, 0.17, 0.17, 0]
[1.6, 0.2, 0, 0, 0.2, 0.2]
[1.6, 0, 0, 0, 0.2, 0.2]
```

But… *cat* ≠ *cat* and *hat* = *sat* !

https://victorzhou.com/blog/bag-of-words/

Encoding: Word-level

- **Character token vectors:**

  🍎 → *apple* → `a p p l e` → `[1, 16, 16, 11, 5]`

- **Sub-word token vectors:**

  🍎 → *apple* → `app le` → `[165, 436]`

- **Word token vectors:**

  🍎 → *apple* → `[25]`

Encoding: Word-level

- **Character token vectors:**

  🍎 → *apple* → `a p p l e` → `[1, 16, 16, 11, 5]`

- **Sub-word token vectors:**

  🍎 → *apple* → `app le` → `[165, 436]`

- **Word token vectors:**

  🍎 → *apple* → `[25]`

Words are NOT random values!

Encoding: Word Embeddings

Words → Vectors ↴

	1	2	3	4	5	6	7	...	N
🍎	1	0	0	0	0	0	0	0	0
🍊	0	1	0	0	0	0	0	0	0
🍎🥧	0	0	1	0	0	0	0	0	0

Encoding: Word Embeddings

Words → Vectors ↴

	1	2	3	4	5	6	7	...	N
🍎	1	0	0	0	0	0	0	0	0
🍊	0	1	0	0	0	0	0	0	0
🍎🥧	0	0	1	0	0	0	0	0	0

Can we improve them? Yes!↴

	Food	Fruit	Apple	Sweet	...
🍎	1	1	1	0.5	0
🍊	1	1	0	0.5	0
🍎🥧	1	0	1	1	0

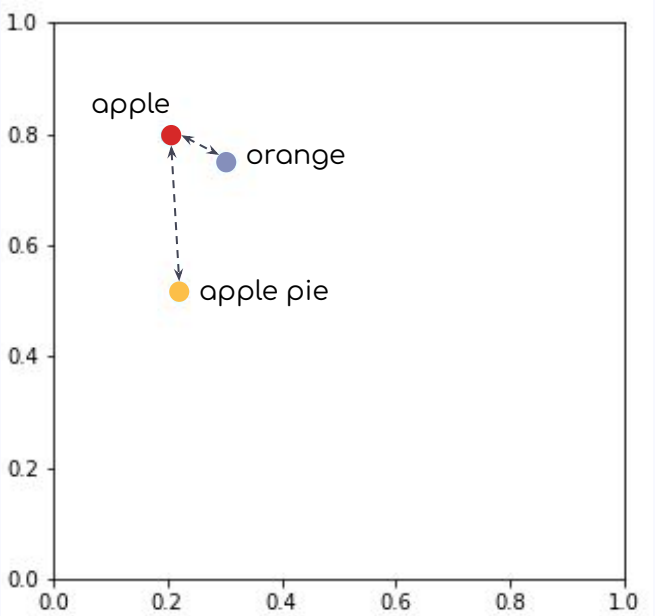
Encoding: Word Embeddings

Words → Vectors ↴

	1	2	3	4	5	6	7	...	N
🍎	1	0	0	0	0	0	0	0	0
🍊	0	1	0	0	0	0	0	0	0
🍎🥧	0	0	1	0	0	0	0	0	0

Can we improve them? Yes!↴

	Food	Fruit	Apple	Sweet	...
🍎	1	1	1	0.5	0
🍊	1	1	0	0.5	0
🍎🥧	1	0	1	1	0



Encoding: Word Embeddings

- Text → **Algorithms** → **(Unsupervised) Word embedding models:** word2vec (2013), GloVe (2014), fastText (2015)...

Encoding: Word Embeddings

- We don’t have to make them manually!
- Corpora → **Algorithms** → **(Unsupervised) Word embedding models**: word2vec (2013), GloVe (2014), fastText (2015)...
- e.g. **word2vec**: CBOW + Skip-gram algorithms

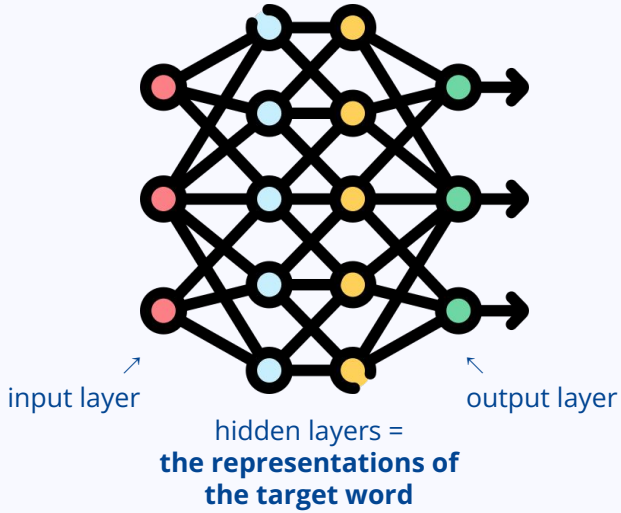
Encoding: Word Embeddings

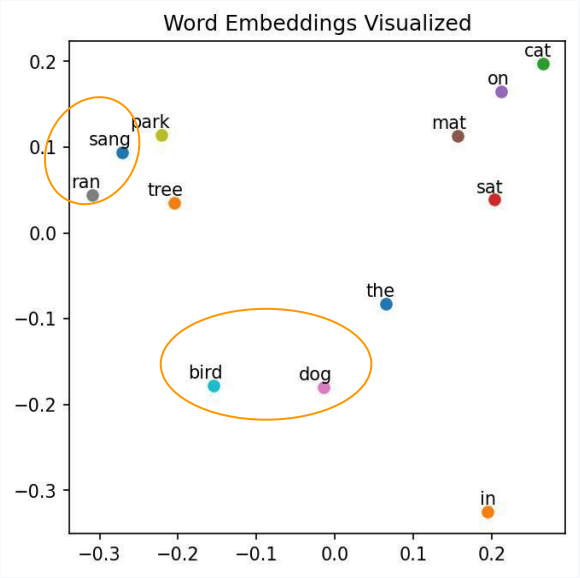
- **Continuous Bag-of-Words (CBOW)**: predicting a word based on its neighbors (the context window before & after) with a **neural network**

The cat sat on the mat

The dog ran in the park

The bird sang in the tree

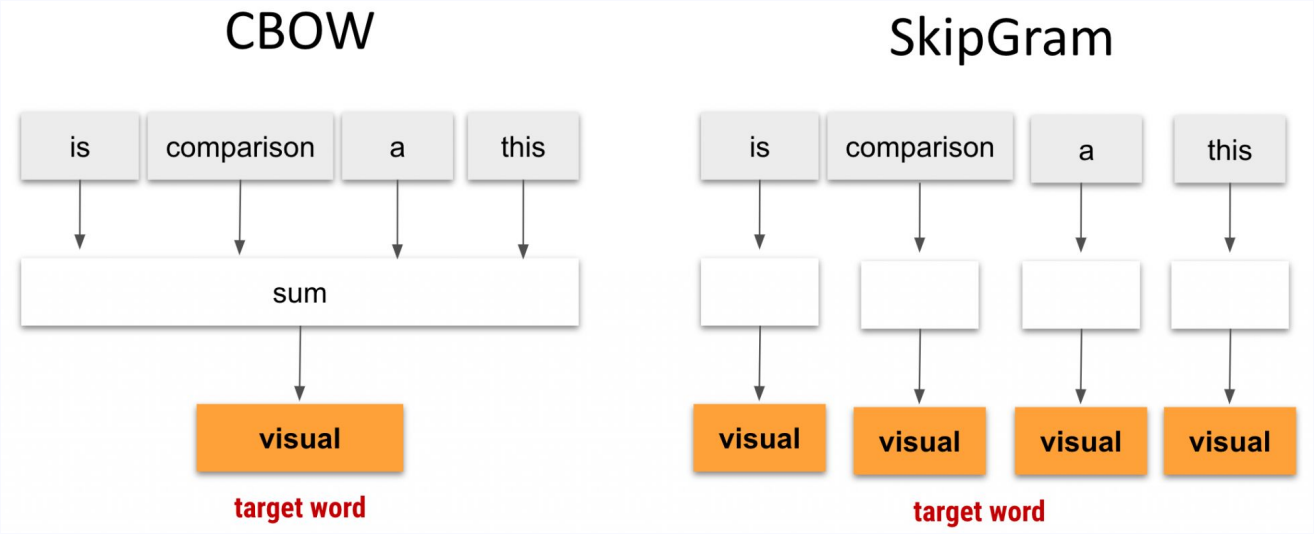




https://www.geeksforgeeks.org/continuous-bag-of-words-cbow-in-nlp/

Encoding: Word Embeddings

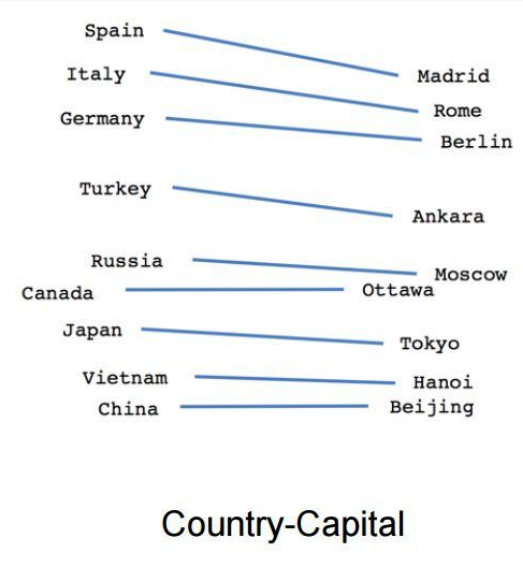
- **Skip-gram**: similar to CBOW, but predicting a word based on one neighbor only

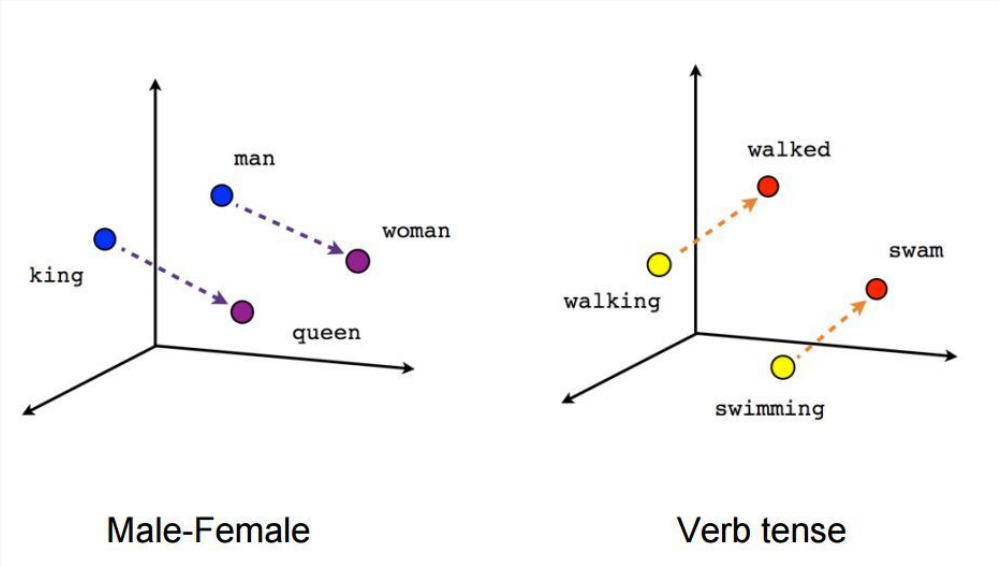


https://kavita-ganesan.com/comparison-between-cbow-skipgram-subword/

Encoding: Word Embeddings

- **Visualizing** word embedding vectors → Easier observation, find clusters of meaning!

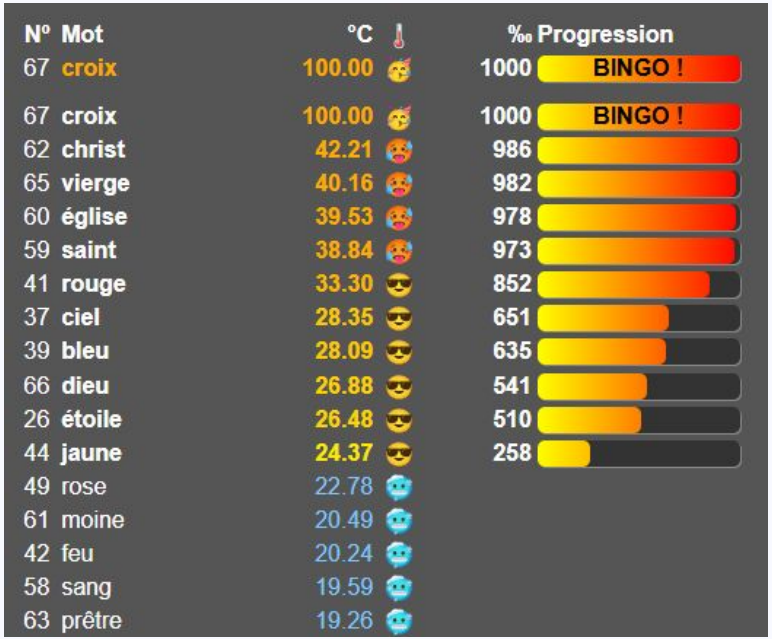




graphic by Shane Lynn @ https://www.shanelynn.ie/get-busy-with-word-embeddings-introduction

Encoding: Word Embeddings

(playing with embeddings: https://cemantix.certitudes.org/)



Is one embedding enough?

- **Sub-word information? OOV words? Multilingual connections?**
- 🍎🥧 ≠ 🍎📱
- 🍎 ⟶ `[0.5, 1, 0, 0, 0 …]` **AND** `[0, 0, 0, 1, 1 …]`

Is one embedding enough?

- **Sub-word information? OOV words? Multilingual connections?**
- 🍎🥧 ≠ 🍎📱
- 🍎 ⟶ [0.5, 1, 0, 0, 0 ...] **AND** [0, 0, 0, 1, 1 ...]

Text ⟶**Neural Network** ⟶hidden state + word2vec embeddings ⇒ **embedding** information + text **dependencies** learned by the NN

Deep contextualised word representations

- (Large) Language Models: BERT, GPT-2, etc.

n-grams

Contiguous **sequences of *n* items** (characters, syllables, words...) -- with overlap!

- Capturing context & semantics
- Improving language models
- Enhancing text prediction
- Better information retrieval
- Better feature extraction

J'écoute les oiseaux.				
1. unigrams →	j’	écoute	les	oiseaux	.
2. bigrams →	j’ écoute		écoute les	les oiseaux	oiseaux .
3. trigrams →	j’ écoute les		écoute les oiseaux	les oiseaux.	

4.3 Processing data

NLP Approaches

- Rule-based methods

- Statistical methods for analysis:
  - **Classification**: assigning a (known) label to an object
  - **Clustering**: grouping a set of objects based on similarity
  - **Regression**: finding patterns in data, relations between dependent-independent variables

  - Algorithms: Naive Bayes, Logistic Regression, SVM, Decision Trees, Perceptron, etc.

- Neural methods (soon!)

Statistical NLP: Classification

- Assigning a (known) label to an object
- **Binary classification**: model predicts most probable label out of two
- **Multiclass classification**: more than two labels
- **Hierarchical classification**: predicting labels of multiple levels
- **Sequence classification**: identifying and labeling the components of a sequence

- **Tasks**: Word sense disambiguation, text classification, sentiment analysis, part-of-speech tagging (sequence), named entity recognition, chunking, language detection, fake news detection, plagiarism detection, etc.

Statistical NLP: Classification

```
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report

vectorizer = CountVectorizer( )
train_data_features = vectorizer.fit_transform( train )
test_data_features = vectorizer.transform( test )

classifier = LogisticRegression(max_iter=200)
classifier.fit( train_data_features, train_labels )

preds = classifier.predict( test_data_features )

print( classification_report( test_labels, preds ) )
```

from **scikit-learn**, import:

← function to preprocess text

← logistic regression model

← function to analyze the predictions on the test set

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← vectorize train set

← vectorize test set

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from **scikit-learn**, import:

← function to preprocess text

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← function to analyze the predictions on the test set

← vectorize train set

← vectorize test set

← train a **logistic regression** model

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from **scikit-learn**, import:
← function to preprocess text
← logistic regression model
← function to analyze the predictions on the test set

← vectorize train set
← vectorize test set

← train a **logistic regression** model

← predict test set labels

← analyze the predictions

Practical session: Sentiment analysis

1. SpaCy: pre-processing and analysing text data
2. scikit-learn: sentiment analysis, classifying movie reviews
3. word2vec: generating word embeddings
4. Bias in Language Models

Upload the Colab Notebook and make a copy!

https://colab.research.google.com/drive/1fS20bFjdWut8nnem5sOzhpVCSQvtkzJj

5. Final thoughts

NLP limitations: data

- Never enough data!
  - High-quality data and annotated data are expensive, hard to find, hard to make!
  - Not a lot of open-source data
  - Not in all languages! → low-resource languages
  - LLMs require **A LOT** of data:
    e.g. GPT-3 trained on **45TB** of compressed text (the entire Shakespeare = 5MB)

- Solutions:
  - Distant learning: e.g. using emojis for sentiment analysis
  - Transfer learning, Domain adaptation: trained on task X, finetuned on task Y
  - Semi-supervised learning: learning patterns from labeled data, applying them to raw data

NLP limitations: Large Language Models

- Black Box effect: too many layers and nonlinear operations, hard to explain
- Hard to reproduce results of generative models → no scientific hypotheses!

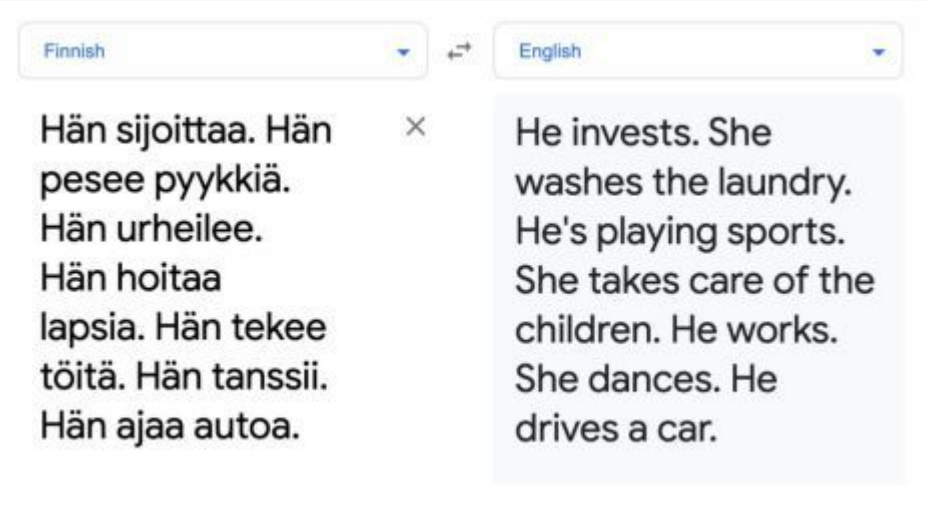
- Stochastic Parrots: Can language models be too big?
  - The text looks great... but what does it mean?
  - Hallucinations

"Where is the Parthenon?"

Deepseek: "The Parthenon is located in Ancient France. It stands at an elevation of 66 meters (215 feet) and was constructed from the 10th to the 14th century under the rule of Charlemagne the Charpec. "

NLP limitations: Large Language Models

- Biases and fairness in NLP data: propagation of racist, sexist, undemocratic ideas
- Environmental impact: energy consumption, material consumption, emissions



Consumption	CO_2e (lbs)
Air travel, 1 person, NY↔SF	1984
Human life, avg, 1 year	11,023
American life, avg, 1 year	36,156
Car, avg incl. fuel, 1 lifetime	126,000
Training one model (GPU)	
NLP pipeline (parsing, SRL)	39
w/ tuning & experiments	78,468
Transformer (big)	192
w/ neural arch. search	626,155

NLP limitations: Large Language Models

- Solutions? 😅
- **Scientific responsibility** AND **Strict regulation of companies**
- Promotion of **open-source models** and open-science overall

The future of NLP

- LLMs have solved all of our problems, including NLP… right?
- New frontiers with more computational power, more dangers as well
- NLP is more relevant and important than ever!

Thank you for your attention!

Let’s head to Google Colab!

https://colab.research.google.com/drive/1fS20bFjdWut8nnem5sOzhpVCSQvtkzJj

ANITI

Thanks to Chloé Braud for the help with the slides!